

Computer Networks Exam Reconstruction

June 10, 2025

Closed Book Questions

1. Layering and the Application Layer

- **Layering**
 - Draw the OSI, TCP/IP, and Tanenbaum protocol stacks side by side.
 - Specify one protocol corresponding to each layer of the Tanenbaum stack.
- **Gnutella 0.4**
 - Describe the function and use of the following message types: PING, PONG, QUERY, and QUERY HIT. Include their routing mechanisms.
 - Explain what happens when a node or file is discovered.
 - Describe how message propagation is terminated.
- **DNS**
 - Justify the use of UDP rather than TCP for DNS queries.

2. Transport Layer

- **Tinygram and Silly Window Syndrome**
 - Define the tinygram problem and discuss common solutions.
 - Explain the silly window syndrome and describe mitigation techniques.
- **QUIC**
 - Does QUIC implement congestion control? Motivate your answer.
 - Compare the congestion control mechanisms of QUIC to those of TCP Reno and TCP Tahoe.
- **ECN**
 - Describe the operational principles of ECN.
 - Compare ECN to RED in terms of benefits and drawbacks.

Open Book Questions

3. Network Layer

– IP Addressing and Subnetting

- Given the network 152.16.168.196/28, determine whether the IP address 162.16.168.30 is within the same subnet.
- Explain the meaning of the CIDR notation /28.
- What does the subnet mask 255.255.240.0 indicate regarding the number of available hosts?
- Determine the broadcast address of the network 192.168.242.5/20.

– IPv6 Address Configuration

- Compare the advantages and disadvantages of IPv6 address autoconfiguration versus DHCP-based assignment.

– Network Address Translation (NAT)

- Discuss problems introduced by NAT at the application layer.
- Provide two examples of techniques used by applications to overcome this problem.

4. Data Link and Physical Layer

– B-MAC Protocol

- Give a situation where you would choose B-MAC over TSMP. Motivate your answer.
- How should B-MAC be configured in scenarios with high vs. low message frequency?

– Sliding Window: Stop-and-Wait

- Identify a type of communication medium where Stop-and-Wait performs well.
- Identify a type of communication medium where Stop-and-Wait performs poorly.
- Given a specific medium, its RTT, and a feasible packet size, estimate the maximum achievable throughput using Stop-and-Wait.